Ankur Barsode

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Postdoctoral Fellow (gravitational-wave astronomy) at the Chennai Mathematical Institute

PhD thesis submitted to the International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru (August 2026), titled "Strong Gravitational Lensing of Gravitational Waves"

MSc (Physics), Birla Institute of Technology and Science Pilani, Rajasthan, 2021

Research Interests

Detecting gravitational lensing of gravitational waves (GWs), astrophysics and cosmology using GWs, Bayesian statistics and data analysis

Experience

- Postdoctoral Fellow, Chennai Mathematical Institute, Chennai (2026–present)
- PhD Student, ICTS-TIFR, Bengaluru (2021–2026)
- Short-term Visitor, Chinese University of Hong Kong (CUHK), Hong Kong (2026)
- Short-term Visitor, Albert Einstein Institute (AEI), Potsdam (2024)
- Short-term Visitor, UCLouvain, Belgium (2024)
- Long-term Research Student, National Center for Radio Astronomy, Pune (2021)
- Summer Research Student, National Center for Radio Astronomy, Pune (2020)

LIGO–Virgo–KAGRA (LVK)

- Data analyst: search for strongly lensed GWs (2022–present)
- Editor: LVK Collaboration paper on the search for lensed GWs in O4b and O4c (2025–present)

Honors and Awards

- International Travel Support from ANRF for presenting my work at the Summer School in Cosmology at the International Centre for Theoretical Physics (ICTP), Trieste (2024)
- Prabhat Award by BITS alumnus to the best outgoing research-oriented student of physics (2021)
- KVPY by the Department of Science and Technology, Government of India, to encourage research at the undergraduate level (2016)

Talks and Presentations

Invited Talks and Seminars

- "Detecting strongly lensed gravitational waves," CUHK, Hong Kong (2026)
- "Detecting strongly lensed gravitational waves," Indian Institute of Science (IISc), India (2026)
- "GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures," LVK Webinar (2026)
- "PO2.0: Fast and efficient GW strong lensing search," AEI, Potsdam (2024)
- "PO2.0: Fast and efficient GW strong lensing search," UCLouvain (2024)

Contributed Talks and Posters

- "Gravitational wave chirp bottle," LVK September meeting, Inter-University Center for Astronomy and Astrophysics (IUCAA), Pune (2026)
- "Detection prospects of strongly lensed gravitational waves," Multi-Messenger Astrophysics in the Dynamic Universe, Yukawa Institute of Theoretical Physics, Kyoto (2026)
- "Detection prospects of strongly lensed gravitational waves," The Future of Gravitational-Wave Astronomy, ICTS, Bengaluru (2025)

– "Detecting a population of strongly lensed gravitational waves," Astronomical Society of India meeting, National Institute of Technology, Rourkela (2025)
– "Constraints on compact DM from the non-observation of GW strong lensing," Summer School in Cosmology, ICTP, Trieste (2024)
– "Constraints on compact DM from the non-observation of GW strong lensing," Astronomical Society of India meeting, IISc, Bengaluru (2024)
– "Hybrid methods to simulate the cosmological HI field," Frontiers in Cosmology, Raman Research Institute, Bengaluru (2023)
– "Constraints on compact DM from the non-observation of GW strong lensing," IAGRG, Indian Institute of Science Education and Research, Kolkata (2022)

Teaching and Mentoring

– Lecturer, GW-Open Data Workshop, ICTS-TIFR, Bengaluru (2025–2026)
– Tutor, GW-Open Data Workshop, ICTS-TIFR, Bengaluru (2024–2026)
– Tutor, Summer School on GW astronomy, ICTS-TIFR, Bengaluru (2023–2025)
– Teaching Assistant, Tools of Astrophysics, ICTS-TIFR, Bengaluru (2024)
– Mentoring a PhD student in the basics of doing research (2025–present)
– Guiding a master's student in their thesis and research (2026–present)
– Informally advised a master's student on a research project (2025)

Service and Outreach

– Organizer, GW-Open Data Workshop, ICTS-TIFR, Bengaluru (2024–2025)
– Reviewer, LIGO–Virgo–KAGRA Papers and Publications
– Invented The Gravitational Wave Chirp Bottle to take GW astronomy to households
– Organized and conducted public demonstrations as a member of Astro Club (ICTS-TIFR)
– Contributed to an exhibition on Climate Change (ICTS-TIFR)

Skills

Programming	Python, C++, JavaScript; git, large-scale computing, pipelining
Analysis	gravitational-wave data analysis, statistics, parameter estimation, machine learning
Visualization	Matplotlib, Processing/p5js, Blender
Communication	Scientific writing (LaTeX, Markdown), presentation (slides, posters, blackboard)
Miscellaneous	Arduino, workshop
AI	Efficient, reliable workflows with Agentic AI

Selected Publications

– Barsode, Ankur. "Tool for Accurate Model Selection Using Bayes Factors." *Physical Review D*, vol. 114, 2026, article 064045. [link]
– Barsode, Ankur, Koustav N. Maity, and Parameswaran Ajith. "Search for Strong Lensing of Gravitational Waves in the Binary Black Hole Events from O1–O4a." *The Astrophysical Journal*, vol. 1009, no. 1, 2026, article 43. [link]
– Barsode, Ankur, Koustav N. Maity, and Parameswaran Ajith. "Lensing, Not Luck! Detection Prospects of Strongly Lensed Gravitational Waves." *The Astrophysical Journal*, vol. 998, no. 2, 2026, article 294. [link]

Full publication list can be found at INSPIRE-HEP and Google Scholar.